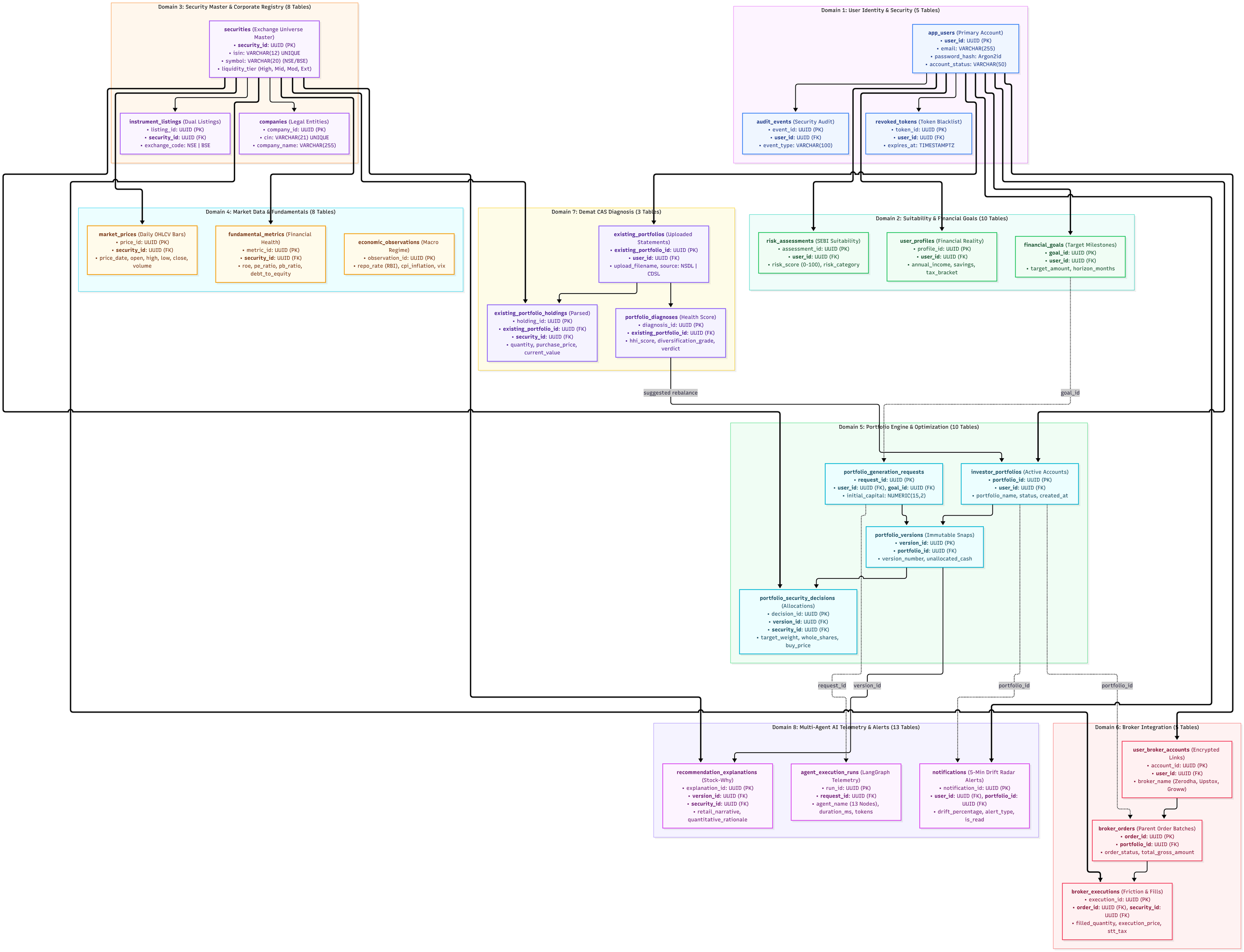

Domain 3: Security Master & Corporate Registry (8 Tables)
securities (Exchange Universe Master)
• security_id: UUID (PK)
• isin: VARCHAR(12) UNIQUE
• symbol: VARCHAR(20) (NSE/BSE)
• liquidity_tier (High, Mid, Mod, Ext)
instrument_listings (Dual Listings)
• listing_id: UUID (PK)
• security_id: UUID (FK)
• exchange_code: NSE | BSE
companies (Legal Entities)
• company_id: UUID (PK)
• cin: VARCHAR(21) UNIQUE
• company_name: VARCHAR(255)
Domain 1: User Identity & Security (5 Tables)
app_users (Primary Account)
• user_id: UUID (PK)
• email: VARCHAR(255)
• password_hash: Argon2id
• account_status: VARCHAR(50)
audit_events (Security Audit)
• event_id: UUID (PK)
• user_id: UUID (FK)
• event_type: VARCHAR(100)
revoked_tokens (Token Blacklist)
• token_id: UUID (PK)
• user_id: UUID (FK)
• expires_at: TIMESTAMPTZ
Domain 4: Market Data & Fundamentals (8 Tables)
market_prices (Daily OHLCV Bars)
• price_id: UUID (PK)
• security_id: UUID (FK)
• price_date, open, high, low, close, volume
fundamental_metrics (Financial Health)
• metric_id: UUID (PK)
• security_id: UUID (FK)
• roe, pe_ratio, pb_ratio, debt_to_equity
economic_observations (Macro Regime)
• observation_id: UUID (PK)
• repo_rate (RBI), cpi_inflation, vix
Domain 7: Demat CAS Diagnosis (3 Tables)
existing_portfolios (Uploaded Statements)
• existing_portfolio_id: UUID (PK)
• user_id: UUID (FK)
• upload_filename, source: NSDL | CDSL
existing_portfolio_holdings (Parsed)
• holding_id: UUID (PK)
• existing_portfolio_id: UUID (FK)
• security_id: UUID (FK)
• quantity, purchase_price, current_value
portfolio_diagnoses (Health Score)
• diagnosis_id: UUID (PK)
• existing_portfolio_id: UUID (FK)
• hhi_score, diversification_grade, verdict
Domain 2: Suitability & Financial Goals (10 Tables)
risk_assessments (SEBI Suitability)
• assessment_id: UUID (PK)
• user_id: UUID (FK)
• risk_score (0-100), risk_category
user_profiles (Financial Reality)
• profile_id: UUID (PK)
• user_id: UUID (FK)
• annual_income, savings, tax_bracket
financial_goals (Target Milestones)
• goal_id: UUID (PK)
• user_id: UUID (FK)
• target_amount, horizon_months
suggested rebalance
goal_id
Domain 5: Portfolio Engine & Optimization (10 Tables)
portfolio_generation_requests
• request_id: UUID (PK)
• user_id: UUID (FK), goal_id: UUID (FK)
• initial_capital: NUMERIC(15,2)
investor_portfolios (Active Accounts)
• portfolio_id: UUID (PK)
• user_id: UUID (FK)
• portfolio_name, status, created_at
portfolio_versions (Immutable Snaps)
• version_id: UUID (PK)
• portfolio_id: UUID (FK)
• version_number, unallocated_cash
portfolio_security_decisions (Allocations)
• decision_id: UUID (PK)
• version_id: UUID (FK)
• security_id: UUID (FK)
• target_weight, whole_shares, buy_price
request_id
version_id
portfolio_id
portfolio_id
Domain 8: Multi-Agent AI Telemetry & Alerts (13 Tables)
recommendation_explanations (Stock-Why)
• explanation_id: UUID (PK)
• version_id: UUID (FK)
• security_id: UUID (FK)
• retail_narrative, quantitative_rationale
agent_execution_runs (LangGraph Telemetry)
• run_id: UUID (PK)
• request_id: UUID (FK)
• agent_name (13 Nodes), duration_ms, tokens
notifications (5-Min Drift Radar Alerts)
• notification_id: UUID (PK)
• user_id: UUID (FK), portfolio_id: UUID (FK)
• drift_percentage, alert_type, is_read
Domain 6: Broker Integration (5 Tables)
user_broker_accounts (Encrypted Links)
• account_id: UUID (PK)
• user_id: UUID (FK)
• broker_name (Zerodha, Upstox, Groww)
broker_orders (Parent Order Batches)
• order_id: UUID (PK)
• portfolio_id: UUID (FK)
• order_status, total_gross_amount
broker_executions (Friction & Fills)
• execution_id: UUID (PK)
• order_id: UUID (FK), security_id: UUID (FK)
• filled_quantity, execution_price, stt_tax